



Mainnet Validation Runbook

cMATRA Token Merger Portal

Prepared for: Flux Point Studios

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Version: 1.0

Classification: Internal Operations

Related: Security Audit Report (April 14, 2026)

PRE-LAUNCH VALIDATION

1. Overview

This document describes the three-level mainnet validation procedure for the cMATRA Token Merger Portal. Each level increases in risk and realism, allowing the team to build confidence that the full surrender pipeline works correctly on mainnet before opening the redemption window to the public.

Level	Test	Risk	What It Proves
1	Dry-run <code>evaluate_tx</code>	Zero	Plutus script executes against live mainnet UTxOs. Correct script hash, valid parameters, fees, UTxO references.
2	Admin canary surrender	Minimal	Full end-to-end pipeline on mainnet: build, co-sign, submit, on-chain confirmation. Verifies cMATRA delivery, legacy quarantine, pool remainder.
3	Portal wallet flow	Minimal	Full user experience: browser portal, CIP-30 wallet connect, scan, confirmation dialog, wallet signature, on-chain submission.

Principle: Each level must pass before advancing to the next. Level 1 failure means "do not touch mainnet." Level 2 failure means "do not open the portal." Level 3 success means "ready for public launch."

2. Prerequisites

The following must be completed before any mainnet validation can begin:

#	Prerequisite	Verification	Status
1	Mint cMATRA on mainnet (1B supply, 6 decimals)	Verify token on Cardanoscan by policy ID	Pending
2	Deploy surrender validator to mainnet (compile with real admin PKH + deadline via <code>aiken blueprint apply</code>)	Verify script address on Cardanoscan	Pending

3	Fund pool UTXOs with cMATRA at script address (Void datums)	Query script address UTXOs via Blockfrost	Pending
4	Configure environment variables on service host	Service <code>/health</code> returns all-green	Pending
5	Security audit completed, all findings remediated	Security Audit Report v1.0	Done
6	Preprod rehearsal passed (9/9 stages)	Preprod rehearsal log	Done

2.1 Required Environment Variables

```
# Python surrender service
ADMIN_SKEY_PATH=/secure/path/to/admin.skey
SURRENDER_API_SECRET=<generate: python -c "import secrets;
print(secrets.token_urlsafe(32))">
SURRENDER_SCRIPT_ADDRESS=addr1...
CMATRA_POLICY_HEX=<cMATRA minting policy hash>
CMATRA_ASSET_HEX=<cMATRA asset name hex>
QUARANTINE_ADDRESS=addr1...

# Next.js (Vercel)
SURRENDER_API_URL=https://<service-host>:8420
SURRENDER_API_SECRET=<same value as above>
BLOCKFROST_PROJECT_ID=<mainnet project ID>
```

3. Level 1 — Dry-Run Evaluation

ZERO RISK

No transaction is submitted to the network.

3.1 What It Does

Builds a complete surrender transaction using real mainnet UTxOs (the funded pool, the user's address), then submits the serialized transaction to Blockfrost's `evaluate_tx` endpoint. This endpoint runs the Plutus V3 script against the live ledger state and returns execution units (CPU + memory) or a detailed error if the script fails. The transaction is never broadcast to the network.

3.2 Procedure

```
# 1. Start the surrender API service
py -3.12 -m services.surrender_api

# 2. In a second terminal, run the canary script (evaluate only)
py -3.12 scripts/mainnet_canary.py \
  --address addr1<your_admin_wallet_address> \
  --asset AGENT \
  --qty 1
```

3.3 Expected Output

```
HEALTH CHECK
  All systems GO

POOL STATUS
  Pool has 3 UTxO(s), 818,659,262.01 cMATRA remaining

LEVEL 1: EVALUATE (dry-run, zero risk)
  Address:  addr1qx...
  Asset:    AGENT
  Quantity: 1
  TX Hash:  a1b2c3d4...
  cMATRA:   0.544576
  Fee:      350000 lovelace (0.3500 ADA)
  Evaluation: {spend:0: ExUnits(mem=..., steps=...)}

LEVEL 1 PASSED — Script evaluation succeeded on mainnet ledger
```

3.4 Pass / Fail Criteria

Outcome	Meaning	Action
PASSED	Script evaluation returned execution units	Proceed to Level 2
FAILED	Script evaluation returned an error	Do NOT proceed. Diagnose: wrong script params? Missing datum? Incorrect pool UTxO format? Fix and re-run.

4. Level 2 — Admin Canary Surrender

MINIMAL RISK

A real transaction is submitted. Uses a trivially small amount.

4.1 What It Does

Performs a real surrender of a tiny amount of legacy assets from the admin's own wallet. The admin key acts as both the service signer and the user signer (self-test). The transaction is submitted to the Cardano mainnet. After on-chain confirmation, the result is verified: cMATRA received, legacy asset quarantined, pool remainder returned.

Cost: Surrendering 1 AGENT costs approximately 0.35 ADA in fees and permanently burns 1 AGENT token (worth fractions of a cent). The admin receives ~0.54 cMATRA in return. This is a real, irreversible transaction.

4.2 Procedure

```
# Run with --submit flag (prompts for confirmation)
py -3.12 scripts/mainnet_canary.py \
  --address addr1<your_admin_wallet_address> \
  --asset AGENT \
  --qty 1 \
  --submit
```

4.3 Post-Submission Verification

After the script reports success, verify on Cardanoscan:

1. **cMATRA output** — The admin address received the correct cMATRA amount
2. **Quarantine output** — 1 AGENT was sent to the quarantine address

3. **Pool remainder** — The remaining pool balance was returned to the script address with a Void datum
4. **Fee** — Transaction fee is reasonable (typically 0.3-0.5 ADA for a Plutus tx)

4.4 Pass / Fail Criteria

Outcome	Meaning	Action
PASSED	All four verification points confirmed on-chain	Proceed to Level 3
FAILED	Submission rejected or outputs incorrect	Do NOT proceed. Investigate the on-chain result or error. The pool UTxO may need to be re-evaluated.

5. Level 3 — Portal Wallet Flow

MINIMAL RISK

Full user-facing flow through the web portal with a real CIP-30 wallet.

5.1 What It Does

A team member (admin or trusted tester) opens the merger portal in a browser, connects a Cardano wallet (Eternl, Lace, Nami, etc.), and completes a surrender through the full user interface. This validates the entire stack end-to-end: frontend rendering, wallet integration, server-side scanning, co-signing flow, wallet signature prompt, and on-chain submission.

5.2 Procedure

1. Navigate to `fluxpointstudios.com/matra-merger`
2. Enter the portal access password
3. Click "Connect Wallet" and select your Cardano wallet
4. Verify the scan results: correct assets and cMATRA amounts displayed
5. Click "Surrender All & Receive cMATRA"
6. Review the confirmation dialog — verify it shows the correct total and irreversibility warning
7. Click "Confirm Surrender"
8. Approve the transaction in your wallet (review inputs/outputs in the wallet's tx preview)
9. Wait for on-chain confirmation
10. Verify the success state: tx hash link displayed, cMATRA in wallet

5.3 Verification Checklist

#	Check	Expected	Result
1	Portal loads behind password gate	Yes	
2	Wallet connects successfully	Yes	
3	Asset scan shows correct balances	Matches wallet	
4	Rate table matches published rates	Matches docs	
5	Confirmation dialog shows amount + warning	Yes	

6	Wallet signature prompt shows correct tx	Review outputs
7	Transaction submitted successfully	Tx hash shown
8	cMATRA arrived in wallet	Correct amount
9	Legacy assets at quarantine address	Cardanoscan
10	Pool remainder at script address	Cardanoscan

6. Launch Readiness Gate

The redemption window may be opened to the public when all of the following are true:

Launch gate criteria:

- ☐ Level 1 passed (evaluate_tx dry-run)
- ☐ Level 2 passed (admin canary surrender verified on-chain)
- ☐ Level 3 passed (full portal flow with wallet co-sign)
- ☐ Security audit — all findings remediated (April 14, 2026)
- ☐ Preprod rehearsal — 9/9 stages passed
- ☐ Rate table published and verified
- ☐ Legacy reward materialization confirmed (or denominator adjusted)
- ☐ Post-deadline policy documented
- ☐ `WINDOW_OPEN = true` set in portal code and deployed

Once all criteria are met, set `WINDOW_OPEN = true` in `pages/matra-merger.tsx`, deploy, and announce the redemption window opening.